

Compound **I** was identified as a novel allosteric inhibitor of Enzyme **X**. The measured inhibitor dissociation constant K_i between the inhibitor and Enzyme **X** was found to be 54 μM without any substrate present. With saturating amounts of nonreactive substrate analog, the measured K_i was found to be 76 μM . Compound **I** is best described as:

- ☐ A. a mixed inhibitor that results in a lower apparent K_m relative to the uninhibited enzyme. [32%]
- ☐ B. a mixed inhibitor that results in the same apparent K_m relative to the uninhibited enzyme. [4%]
- ✔ ☒ C. a mixed inhibitor that results in a higher apparent K_m relative to the uninhibited enzyme. [58%]
- ☐ D. a purely noncompetitive inhibitor that only lowers the apparent $V_{\max}$ relative to the uninhibited enzyme. [4%]

Correct

58% answered correctly

Explanation:

Mixed inhibitors that bind free enzyme more strongly cause an increase in the apparent K_M

$K_{iE} = \infty$ (does not bind E) Only binds ES complex	Uncompetitive	$K_M \downarrow$ (by same % as $V_{\max}$)	$V_{\max} \downarrow$
$K_{iE} > K_{iES}$ Binds ES more strongly than E	Mixed (more uncompetitive-like)	$K_M \downarrow$ (by smaller % than $V_{\max}$)	$V_{\max} \downarrow$
$K_{iE} = K_{iES}$ Binds ES and E equally	Purely noncompetitive	K_M unchanged	$V_{\max} \downarrow$
$K_{iE} < K_{iES}$ Binds E more strongly than ES	Mixed (more competitive-like)	$K_M \uparrow$	$V_{\max} \downarrow$
$K_{iES} = \infty$ (does not bind ES) Only binds free enzyme E	Competitive	$K_M \uparrow$	$V_{\max}$ unchanged

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Reversible enzyme inhibitors can be classified by **which species they bind** and **by their effects on the apparent Michaelis–Menten parameters**. Competitive inhibitors bind only the free enzyme (E) and **increase the apparent K_m value**. In contrast, uncompetitive inhibitors bind only the enzyme–substrate complex (ES) and **decrease both K_m and $V_{\max}$ by the same percent**.

A mixed inhibitor reversibly **binds to both E and ES**, and always lowers the apparent $V_{\max}$; however, its effect on K_m differs based on the relative affinities between the inhibitor and each enzyme species. Purely noncompetitive inhibitors bind both E and ES with **equal affinity** and produce **no change in the apparent K_m** . Inhibitors that bind **more strongly to E** than to ES are mixed inhibitors that behave more like competitive inhibitors than uncompetitive inhibitors, and therefore increase the apparent K_m . In contrast, inhibitors that bind **more strongly to ES** behave more like uncompetitive inhibitors—both $V_{\max}$ and K_m decrease but K_m decreases by a smaller percentage than $V_{\max}$.

In this question, two inhibitor dissociation constants (K_i values) are given. Because the substrate analog mimics the substrate, it induces the enzyme to adopt an ES-like conformation, so the measured K_i value with the analog can be taken as the K_i value for the true ES complex. Given

this, the inhibitor binds the **free enzyme with stronger affinity** (ie, a smaller K_i) than it binds the ES complex (ie, 54 μM vs. 76 μM).

Because **smaller binding constants indicate stronger affinity**, Compound I binds **more strongly to E** than to ES and acts as a **mixed inhibitor that increases the apparent K_m** .

(Choice A) Mixed inhibitors that *lower* the K_m value show stronger affinity toward the ES complex, and would have a *smaller* K_i value when measured with substrate analog.

(Choices B and D) Purely noncompetitive inhibition is a special case of mixed inhibition where the apparent K_m value is unchanged relative to uninhibited enzyme. In these cases, the K_i value would be the same for both the free enzyme and enzyme with substrate analog.

Educational objective:

Mixed inhibitors have different effects on K_m based on their relative affinities to the free enzyme and to the enzyme–substrate complex. Mixed inhibitors that bind more strongly to the free enzyme cause a lower apparent $V_{\max}$ but a higher apparent K_m compared to uninhibited enzyme.

Time spent:0 sec

Subject:
Biochemistry

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System:
Enzymes